

Assignment 1

Course: Deep Learning

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Q5. Find and plot the derivatives of (a) ReLU, (b) Sigmoid, and (c) Hyperbolic Tangent functions.

(a) ReLU

Analytical Form

The Rectified Linear Unit (ReLU) is defined as

$$\text{ReLU}(x) = \max(0, x) = \begin{cases} 0, & x < 0, \\ x, & x \geq 0. \end{cases} \quad (1)$$

Differentiating piecewise (taking the subgradient at $x = 0$ to be 0, as is common in practice) yields

$$\text{ReLU}'(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases} \quad (2)$$

Figure 1 shows the ReLU activation function and its derivative.

Plot

The derivative in python code is simple finite check in the limits given while for the derivative at Zero its taken as 0.

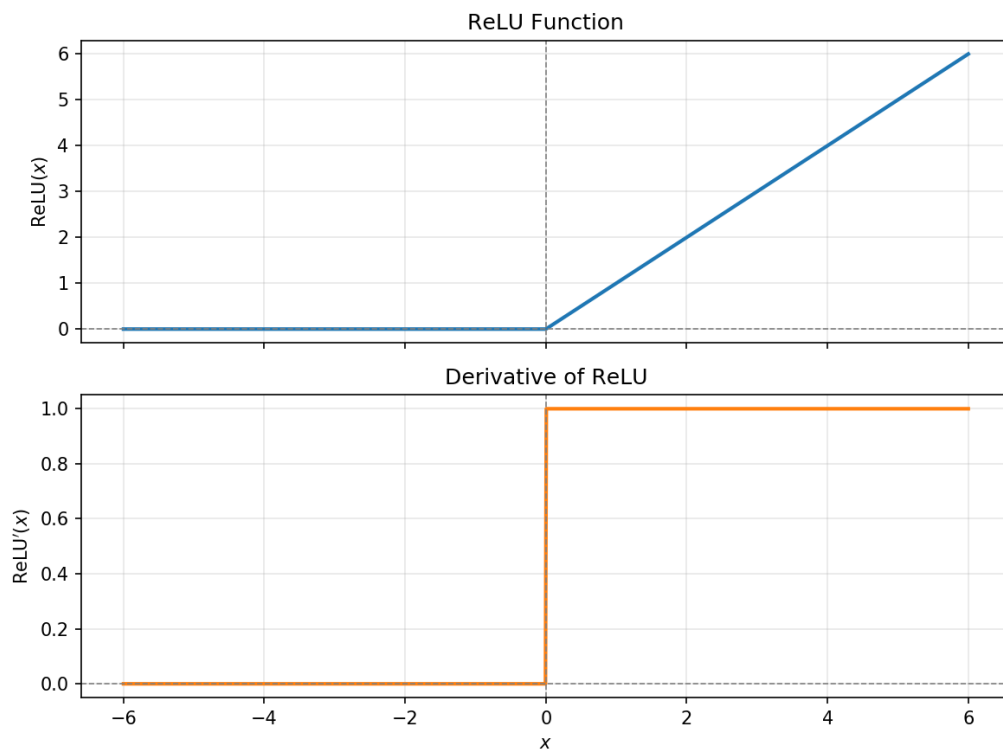


Figure 1: ReLU activation (top) and its derivative (bottom).

Python Implementation

For the computation of derivatives we have used the NumPy library.

```
1 import numpy as np
2 def relu(x):
3     return np.maximum(0.0, x)
4
5 def relu_derivative(x):
6     return (x > 0).astype(float)
```

Listing 1: ReLU and its derivative

(b) Sigmoid

Analytical Form

The logistic sigmoid is

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \quad (3)$$

Differentiating with respect to x :

$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \cdot \frac{e^{-x}}{1 + e^{-x}} = \sigma(x) (1 - \sigma(x)). \quad (4)$$

Thus

$$\sigma'(x) = \sigma(x) (1 - \sigma(x)). \quad (5)$$

Figure

Figure 2 shows the Sigmoid activation function and its derivative. Chain rule has been used to find the derivative.

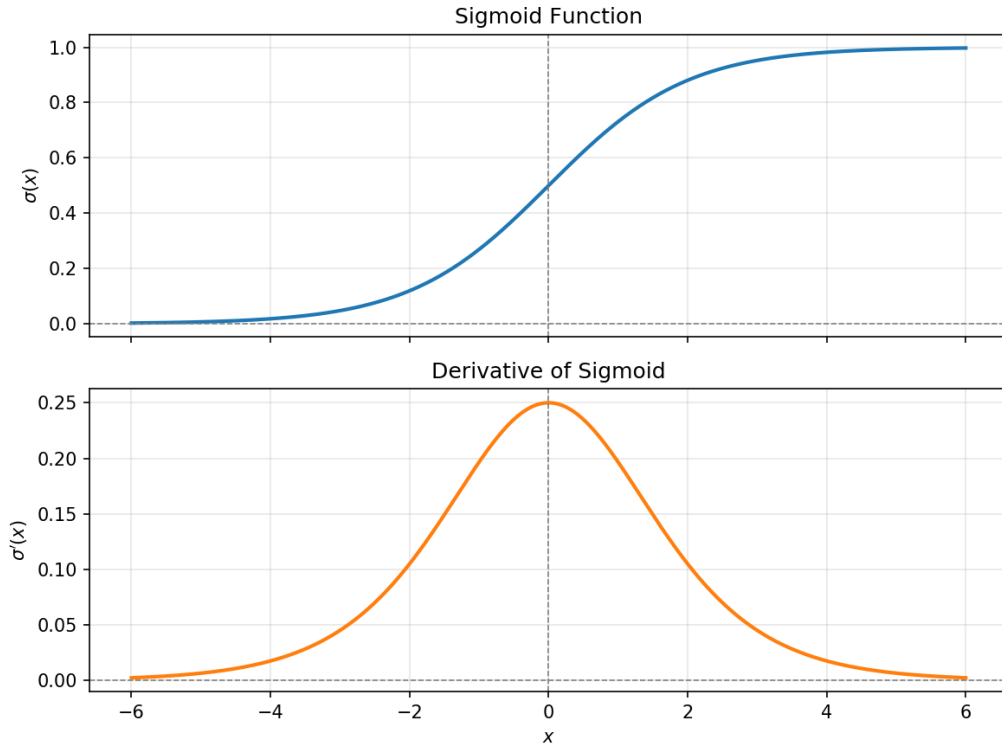


Figure 2: Sigmoid activation (top) and its derivative (bottom).

Python Implementation

```
1 import numpy as np
2
3 def sigmoid(x):
4     return 1.0 / (1.0 + np.exp(-x))
5
6 def sigmoid_derivative(x):
7     s = sigmoid(x)
8     return s * (1.0 - s)
```

Listing 2: Sigmoid and its derivative

(c) Hyperbolic Tangent

Analytical Form

The hyperbolic tangent is

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\sinh(x)}{\cosh(x)}. \quad (6)$$

Its derivative follows from the quotient rule (or the identity $\frac{d}{dx} \tanh x = \text{sech}^2 x$):

$$\tanh'(x) = 1 - \tanh^2(x) = \text{sech}^2(x). \quad (7)$$

Figure

Figure 3 shows the Hyperbolic Tangent activation function and its derivative. Chain rule has been used to find the derivative.

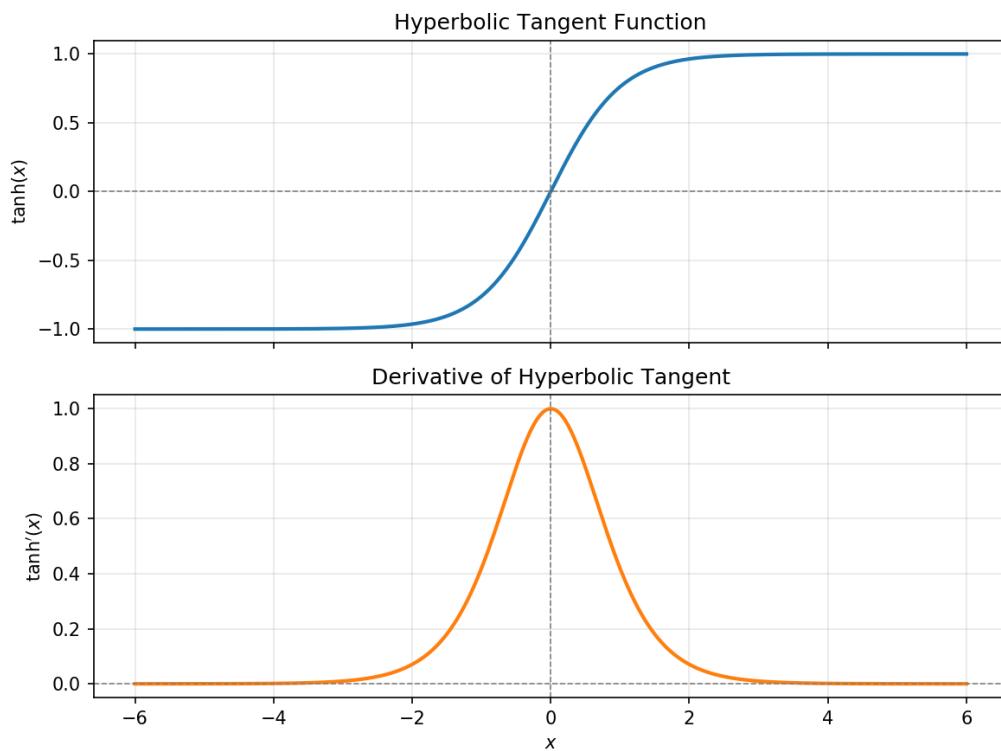


Figure 3: Hyperbolic tangent activation (top) and its derivative (bottom).

Python Implementation

```
1 import numpy as np
2
3 def tanh_fn(x):
4     return np.tanh(x)
5
6 def tanh_derivative(x):
7     t = np.tanh(x)
8     return 1.0 - t**2
```

Listing 3: Hyperbolic tangent and its derivative